MAT1375, Classwork9, Fall2026

$2x^2+3x-2=(x+2)(2x-1)$
$=2(x+2)(x-\frac{1}{2})$
$=2(x-(-2))(x-\frac{1}{2})$

Ch9. Roots of Polynomials

1. Factors and roots of polynomials.

Every n-degree polynomial $f(x)=a_nx^n+a_{n-1}x^{n-1}+\cdots+a_2x^2+a_1x^1+a_0$, $(a_n\neq 0)$ can be factored as

$$f(x)=\underline{\quad a_n(x-c_1)(x-c_2)(x-c_3)(x-c_4)\cdots(x-c_n)\quad}$$

Thus, the polynomial $f(x)$ of degree n has ~~at most~~ exactly $\underline{\quad n\quad}$ roots (which are $c_1, c_2, \cdots, c_n$) and these roots may be either $\underline{\quad \text{real}\quad}$ or $\underline{\quad \text{complex}\quad}$. $(3+2i,\ \sqrt{2}+5i,\ 3+4i)$ $3-2i$ $\sqrt{2}-5i$ $3-4i$

2. The Repeat roots and its **Multiplicity.**

Let $f(x)=(x-r)^n$ where r is the $\underline{\quad \text{root}\quad}$ of f and this root repeats $\underline{\quad n\quad}$ times. We call r a root with $\underline{\quad \text{multiplicity}\quad}$ n.

3. The complex root and its Conjugate.

Let f be a polynomial with all **real coefficients**. The complex roots are always found as a **pair**, that is, if $c=a+bi$ is a complex root of f, then the complex $\underline{\quad \text{conjugate}\quad}$ $\bar{c}=\underline{\quad a-bi\quad}$ is also a root of f. $c=3+5i,\ \bar{c}=3-5i$

4. The Relation between Roots and Coefficient a_0. $f(c_1)=(c_1)^n+a_{n-1}(c_1)^{n-1}+\cdots+a_1c_1+a_0$

For a n-degree polynomial $f(x)=1x^n+a_{n-1}x^{n-1}+\cdots+a_2x^2+a_1x^1+a_0$, we can factorize it as

$$f(x)=1(x-c_1)(x-c_2)(x-c_3)(x-c_4)(x-c_5)\cdots(x-c_n)$$

$f(c_1)=(c_1-c_1)(c_1-c_2)(c_1-c_3)(c_1-c_4)(c_1-c_5)\cdots(c_1-c_n)=0$

where $c_1, c_2, \cdots, c_n$ are the roots of $f(x)$. Then we have

(1) $f(c_1)=\underline{\quad 0\quad}$, $f(c_2)=\underline{\quad 0\quad}$, $f(c_3)=\underline{\quad 0\quad}$, $\ldots$, $f(c_n)=\underline{\quad 0\quad}$.

(2) $a_0=\underline{\quad (-c_1)\cdot(-c_2)\cdot(-c_3)\cdot(-c_4)\cdot\cdots\cdot(-c_n)\quad}$.

(3) The $\underline{\quad \text{factor}\quad}$ of a_0 might be the possible candidates for $\underline{\quad \text{the root}\quad}$ of $f(x)$.

$f(x)=x^3+2x^2+5x+4$
factors of 4 are ± 2, ± 1, ± 4

5. Let $f(x)=x^3-x^2+2$. Find all the roots of $f(x)$.

Step1 guess for roots of $f(x)$ which is the factors of 2: $\pm 1, \pm 2$

Step2 $f(1)=(1)^3-(1)^2+2=1-1+2=2$ ✗

$f(-1)=(-1)^3-(-1)^2+2=-1-1+2=0 \Rightarrow x=-1$ is a root of $f(x)$

$\Rightarrow (x+1)$ is a factor of $f(x)$

Step3 $f(x)=(x+1)(x^2-2x+2)$

$$\begin{array}{r|l} x+1 & x^3-x^2+0x+2 \\ & x^3+x^2 \\ \hline & -2x^2+0x \\ & -2x^2-2x \\ \hline & 2x+2 \\ & 2x+2 \\ \hline & 0 \end{array}$$

$\frac{x^3}{x}=x^2$, $x^2(x+1)$; $\frac{-2x^2}{x}=-2x$, $-2x(x+1)$; $\frac{2x}{x}=+2$

Step4 Find the root of $f(x)$...

$(x+1)(x^2-2x+2)=0 \Rightarrow (x+1)=0$ or $(x^2-2x+2)=0$

$\Rightarrow x=-1$ or $x=\frac{2\pm\sqrt{-4}}{2}$ $\left(ax^2+bx+c=0,\ x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}\right)$

$\Rightarrow x=-1$ or $x=\frac{2\pm 2i}{2}$ $(\sqrt{-4}=\sqrt{4}\cdot\sqrt{-1}=2i)$

$\Rightarrow x=-1$ or $x=\frac{2(1\pm i)}{2}=1\pm i$

3 roots: $x=-1$, $x=1+i$, $x=1-i$

(Real) & (a pair of the complex roots)

6. The **Number Line Test**: Given a polynomial $f(x)$

Step1. Solve the equation $f(x)=0$ and find all the roots.

Step2. Mark the roots on the number line and check sign (positivity/negativity) in each subinterval.

7. Let $f(x) = x^3 - 3x^2 + 4$. Find all the roots of $f(x)$. Sketch a complete graph and label all roots.

① Possible roots: the factors of 4: $\pm 1, \pm 2, \pm 4$

$f(1) = (1)^3 - 3(1)^2 + 4 = 1 - 3 + 4 = 2$ ✗

$f(-1) = (-1)^3 - 3(-1)^2 + 4 = -1 - 3 + 4 = 0$ ✓

$x=-1$ is a root of $f(x)$, $(x+1)$ is a factor of $f(x)$

② long division

$x+1 \;\big|\; x^3 - 3x^2 + 0x + 4$

$x^3 + x^2$

$-4x^2 + 0x$

$-4x^2 - 4x$

$4x + 4$

$4x + 4$

0

③ Find the roots

$f(x) = (x+1)(x^2 - 4x + 4)$

$x \quad -2$

$x \quad -2$

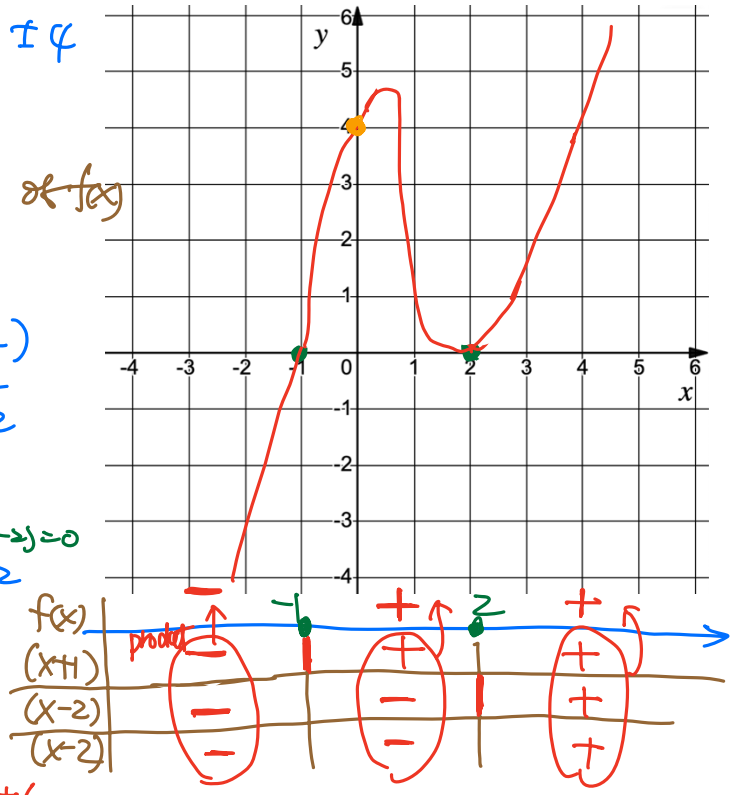
$= (x+1)(x-2)(x-2) = 0$

$\Rightarrow (x+1)=0$ or $(x-2)=0$ or $(x-2)=0$

roots: $x=-1,\ x=2,\ x=2$

④ Number line test

- Mark the roots on number line
- list all factors of $f(x)$
- try the points to check +/-



$3^3 = 27,\ 3^2 = 9$

8. Let $f(x) = x^3 - x^2 - 9x + C$ where C is a real number. If $x = 3$ is a root of $f(x)$, find C so that $f(x)$ has this root as indicated. Then, for this choice of C, find all remaining roots of $f(x)$.

$f(3)=0,\ C=?$

$f(3) = 3^3 - 3^2 - 9\cdot 3 + C = 0$

$\Rightarrow 27 - 9 - 27 + C = 0$

$\Rightarrow -9 + C = 0 \Rightarrow C = 9$

(implies $(x-3)$ is a factor of $f(x)$)

$f(x) = x^3 - x^2 - 9x + 9$

$= x^2(x-1) - 9(x-1)$

$= (x-1)(x^2 - 9)$

$= (x-1)(x+3)(x-3)$

$f(x) = (x-1)(x+3)(x-3) = 0$

$(x-1)=0$ or $x+3=0$ or $x-3=0$

$x=1$ or $x=-3$ or $x=3$

$x-3 \;\big|\; x^2 + 0x - 9$

9. Find a polynomial $f(x)$ that fits the given data.

$f(x)$ has degree 4. $f(x)$ has roots $0, 2, -1, -4$, and $f(1) = 20$.

The factors of $f(x)$ are

$x \quad (x-2) \quad (x+1) \quad (x+4)$ ($x=-1$)

$f(x) = -2 \cdot x(x-2)(x+1)(x+4)$

$f(1) = -2 \cdot 1 \cdot (1-2)(1+1)(1+4)$

$= -2 \cdot (-1)(2)(5) = -2(-10) = 20$